

Bridging Modeling and Domain Expertise Through Visualization: A Case Study on Bread-Making with Bayesian Networks

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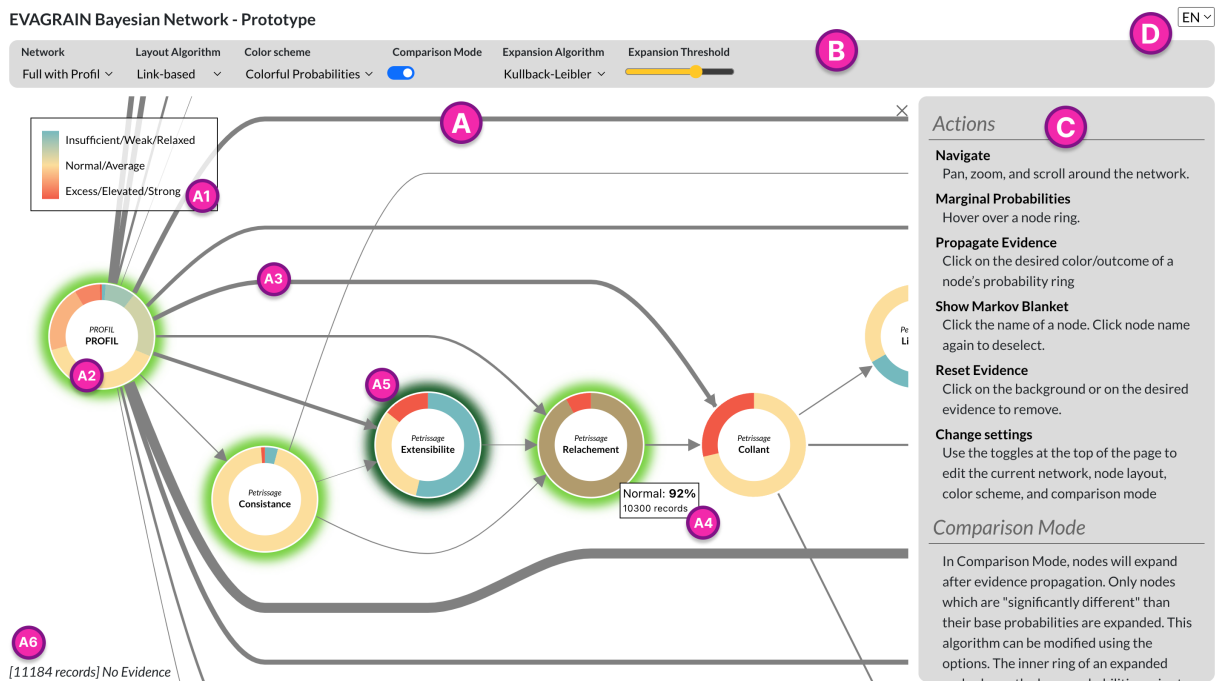


Figure 1: Visualization prototype for exploring a Bayesian network for a bread-making model: Consists of network view (A), options bar (B), help menu (C), and language toggle (D). Within network view, a legend (A1) in the corner serves as a reference for node colors. Nodes (A2) are circles with colored probability arcs and relations (A3) are directed arrows. Hovering over a node arc shows an informative tooltip (A4). Clicking on a node center shows its Markov Boundary (A5). An observation count and evidence list are written in the bottom corner (A6).

Abstract

Decision support tools based on Bayesian Networks (BN) are capable of simply representing complex relationships within data. We use a BN network to model bread-dough behavior, reflecting the bread-making process. Such a network must be validated using domain knowledge, but domain experts often lack the necessary statistical backgrounds to understand BNs. We propose a visualization to enable domain experts without technical background to explain and critically analyze the network. Our platform uses familiar visualizations and focuses on interactive evidence propagation, view customization, and access to the underlying dataset as key features to facilitate understanding. Design workshops and a preliminary evaluation with domain experts and BN specialists showed its potential for exploring and validating the model. We report on lessons learned in creating a visualization for making complex models accessible to domain experts, and how its design can foster interdisciplinary dialogue between modelers and domain experts.

CCS Concepts

• **Human-centered computing** → **Empirical studies in visualization**; **Visual analytics**.

Keywords

visualization, working with domain experts, Bayesian networks, modeling, bread-making

ACM Reference Format:

Omi H Johnson, Melanie Munch, Kamal Kansou, Cedric Baudrit, Anastasia Bezerianos, and Nadia Boukhelifa. 2025. Bridging Modeling and Domain Expertise Through Visualization: A Case Study on Bread-Making with Bayesian Networks. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720102>

1 Introduction

Machine learning models for decision support are increasingly commonplace in many fields. For example, the national EVAGRAIN project [1] aims to model bread-making to create decision support tools for millers and bakers to classify French wheat. In France, wheat grading relies on sensory evaluations of dough and bread from a standard bread-making assay. This sensory data is used to model the behavior of the dough, reflecting the bread-making process and balancing bread quality and societal demands for healthy, sustainable wheat production [20]. This project models this process using a Bayesian Network (BN), a probabilistic Machine Learning (ML) approach that generates scalable models under uncertainty [23]. The EVAGRAIN model was trained on a large bread-making dataset and input from domain experts [21].

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720102>

A crucial step after constructing these models is their validation by domain experts, in our case researchers or practitioners who are familiar with the bread-making process but are not knowledgeable about Bayesian Networks. While Bayesian Networks and other ML methods have powerful modeling capabilities for complex data, they often lack ‘explainability’ for domain experts: expressing the logic behind the network so that experts can understand predictions and significant flows of information. Explainability is critical to establish trust in a BN’s predictions and effectively validate the model [16], particularly in high-risk domains such as the medical field [30]. Non-technical users like domain experts may struggle with concepts like conditional probability tables, which grow exponentially in size for complex networks, or with distinguishing causal and associative relationships [28].

Visualization can summarize complex data and probabilities like those seen in Bayesian Networks [27]. However, existing visualization platforms for Bayesian Networks are either too complex for domain experts (e.g. GENie [13], BayesSuites [17]) or are primarily targeted for medical clinicians and lack context useful for EVAGRAIN’s uniquely data-knowledgeable domain experts (e.g. [12], [18]).

We followed a user-centered design process to develop a prototype visualization platform for the EVAGRAIN Bayesian Network featuring animated, one-click evidence propagation, as a key tool for explainability, to aid domain experts understand, reason about, and validate the relations within the model. The evaluation of our prototype with domain and modeling experts shows that visualization not only makes complex BN models more accessible but also facilitates the dialogue between domain and modeling experts.

2 Related Work

A Bayesian Network (BN) is a probabilistic model based on a directed acyclic graph [23]. BNs can represent complex relationships between variables and are common in expert reasoning systems. Bayesian Networks consist of a node set, a set of directed relations between nodes, and conditional probability tables (CPTs). Nodes represent variables of interest with a set of possible states and likelihoods, termed ‘marginal probabilities’. Directed relations represent causal or associative dependencies between variables. CPTs quantify the probabilistic contribution of parent variables to the likelihood of a child. ‘Evidence’ can be entered into the network by specifying a set of known values, which can then be used to infer the conditional probabilities of other variables in the network (‘Evidence propagation’).

Related work on Bayesian Network visualization, introduces visualizations as: auxiliary support to a model, as a platform for BN development, and more relevant to our work as a platform designed specifically for usability and validation by domain experts. Most BN-focused papers are in the medical field and use a common strategy of representing BNs as directed acyclic graphs with variables as nodes and relations as directed arrows.

Studies focused on developing a usable tool for domain experts, omit showing conditional probability tables directly because of their complexity [12, 15, 18, 19, 26]. Two [15, 26] go so far as to hide the presentation of the network structure itself, instead favoring views that only show variable names and their current probabilities.

This work emphasizes evidence-propagation based exploration, arguing that experts can better observe the relationships between nodes through propagation interactions than complex CPTs [26]. Even expert-focused platforms that retain some CPT presentation [19], use a data-based algorithm to only show uncertain portions of select CPTs.

When representing marginal probabilities, Champion and Elkan [7] propose portraying nodes as pie charts where each slice represents a state's probability. This presentation is digestible to experts who may struggle with traditional bar charts. Evidence comparison is possible in this model by adding another ring of probabilities with a second evidence set. Müller et al. [18] use this approach in their medical BN validation platform, including additional information like the direction of change. Cypko et al. [12] use a similar visual representation, but also introduce color to highlight 'significant' changes in probability. We were inspired by these techniques for our own platform.

Of this work, few directly evaluated their platform using domain experts [12, 18, 19]. All three used a guided, task-based format where participants completed specific actions to assess usability. The three studies all reported positive outcomes, for example Müller et al. [18] reported 33% faster completion of patient assessment when using its clinical decision support BN platform. However, none of these evaluations focused specifically on BN validation.

None of the existing approaches are suited exactly to the needs of the bread-making process. This past work [12, 18] often chooses to simplify the model by only showing a section of the network at a time, while our experts are interested in all variables or all those within a specific bread-making step. Other work addressed this need by using Markov boundaries [6, 8] or fish-eye zooming [25] to examine specific nodes while maintaining context.

3 Domain Context, Motivation and Process

Bread-making is a complex process, influenced by the raw materials and by decisions on how to mix and develop the gluten structure in the dough. Projects such as EVAGRAIN [1], attempt to model this process, incorporating both measurements and expert knowledge. Our goal is to make such models accessible to domain experts and validate them before they can be used in practice by a bigger audience (millers, researchers in bread-making, standards boards).

Our starting point is an existing Bayesian network [1] learned from a dataset of 10,898 bread-making trials from 2003 to 2022. The network models each bread-making step and records dough properties like color and stickiness. The six steps—kneading, pointing, shaping, conditioning, baking, evaluation—include 23 sensory and 2 quantitative measures (volume, hydration). To simplify, sensory measures were grouped into three categories: 'Insufficient', 'Normal', and 'Excess', based on the French bread standard. Quantitative measures were binned into 'Weak', 'Average', and 'High'. An 'PROFIL' variable ties the other measures to a kneading profile summarizing flour quality and, by extension, wheat quality, calculated on a 7-value scale from 'Slackening' to 'Normal' to 'Resistant' [20]. Expert interviews identified variable dependencies and informed constraints to exclude incorrect relations and orient arcs based on the temporal order of bread-making steps. The Bayesian Network

was built using pyAgrum [14] with the Greedy Hill Climbing algorithm guided by BIC score [21]. Two network versions are available: with or without the PROFIL variable.

The modeling experts were either familiar with Bayesian networks or had participated in constructing the EVAGRAIN dataset and Bayesian network, as well as in interviews with domain experts.

4 Understanding Visualization Needs: Design Workshops

To identify user needs we conducted two design sessions with one domain expert and three BN modeling experts. The domain expert was a wheat baking quality researcher with over 10 years of experience. The modeling experts were either familiar with Bayesian networks or had taken part in constructing the EVAGRAIN dataset and Bayesian network, as well as in interviews with domain experts. These modeling experts acted as "proxies" of our domain experts to gain an understanding of possible needs before reaching out to domain experts working in industry. The two sessions lasted 1.5h each (two experts per session).

Procedure: Each session began with a brief review of common BN visualization techniques as inspiration. Participants then sketched an "ideal" Bayesian Network visualization from a domain expert's perspective, noting the potential tasks to support. Suggestions were coded, discussed, and organized using thematic analysis [5] to identify recurring themes.

Outcomes: Participants noted that domain experts would not understand BNs or CPTs, as they think in terms of qualitative interactions between variables. Bread-making experts may focus on specific process steps, making network simplification essential. Suggestions included using pie-charts [7], hiding "unnecessary" nodes and edges, and hiding complex CPTs. There were also significant concerns about experts assuming causality, as it is "very difficult" for experts to think purely in correlative terms. Participants emphasized that domain experts prioritize understanding node and relation influences, focusing on specific interactions or identifying key nodes affecting a particular outcome after evidence propagation. Based on these remarks, we created a list of questions to address with our visualization:

- Q1: How can we reduce the complexity of the EVAGRAIN BN network and conditional probabilities?
- Q2: How can we convey the most significant flow of information between some or all variables?
- Q3: What are the variables that most influence a particular node or outcome?
- Q4: How does the underlying dataset inform node probabilities and relations?
- Q5: How can we avoid assumptions of causality?

5 Implementation

The visualization platform (Figure 1) includes a main network view (A), an options bar (B) with several customization options spanning the top portion of the screen, and a collapsible help panel (C) that is accessible to the right of the network view. Finally, a language toggle allows user to switch between the supported languages of French and English (D). We implemented this platform as an interactive client-server web application using React, JavaScript, D3.js [2],

Python, and the Flask framework. We used the *pyAgrum* python library [14] to load the Bayesian Network files and perform analyses such as inference, mutual information, and Markov Boundaries. Our visualization platform can be easily adapted to other domains, such as healthcare and manufacturing, as it relies on the Bayesian Interchange Format (.bif) to describe the BN model. Currently, it supports models with up to seven discrete values to ensure circle readability.

5.1 Network View

The network view presents the current Bayesian Network. The network consists of 25 circular nodes (26 for the network with the PROFIL node) connected by directed arrows representing relations. The user can pan and zoom around the network view. A legend is placed in the top left corner (Figure 1.A1) with a slight transparency for network visibility. Within the network, the user may perform evidence propagation and highlight Markov Boundaries (discussed next).

Nodes: Each node is a circle (Figure 1.A2) with the variable name and bread-making step displayed at its center. Around the center is a ‘probability ring’, which consists of a pie-chart of arcs representing state probabilities. All variables in the EVAGRAIN BN have the same set of three possible outcomes: Insufficient/Weak rendered as blue, Normal/Average as beige-yellow, and Excess/High as red, except for The PROFIL variable which has 7 values, shown using an interpolated version of the blue/beige/red color scheme (Figure 1.A2). Using a pie chart simplifies the presentation of probabilities while maintaining accurate assessment of relative likelihood (Q1). Hovering over an arc section displays the numerical probability, allowing optional additional detail (Figure 1.A4).

Relations: Relations are represented by directed arrows (Figure 1.A3). To convey a sense of importance (Q2), the thickness of a relation is proportional to its **mutual information**. Mutual information measures information that one random variable contains over another [11]. Mutual information is conditional on the current evidence set and updates after each evidence propagation. If evidence is propagated to a variable, it is no longer random and mutual information is 0; any relations connected to the evidence are shown as a dashed line. This solution addresses Q1 and Q2 by centering visual focus on the thicker, “more important” relations. We support two network layouts: The Link-based layout uses the Sugiyama algorithm [22] to order the nodes in a way that minimizes edge crossings (Figure 1). The step-based layout places nodes in vertical lines according to their bread-making step, ordering steps left to right (Figure 2b).

5.2 Markov Boundaries

To identify connected nodes and influential relationships (Q2), we allow experts to highlight the Markov Boundary (MB) of a target node. The MB of a node is the minimal set of other nodes (parents, children, and children’s parents) that determine the value of the target [6]. By clicking on the center of a node of interest, the node and its MB are highlighted by colored halos (Figure 1.A5). The MB persists until the target’s center is clicked again; this allows the user to maintain the boundary’s context during evidence propagation.

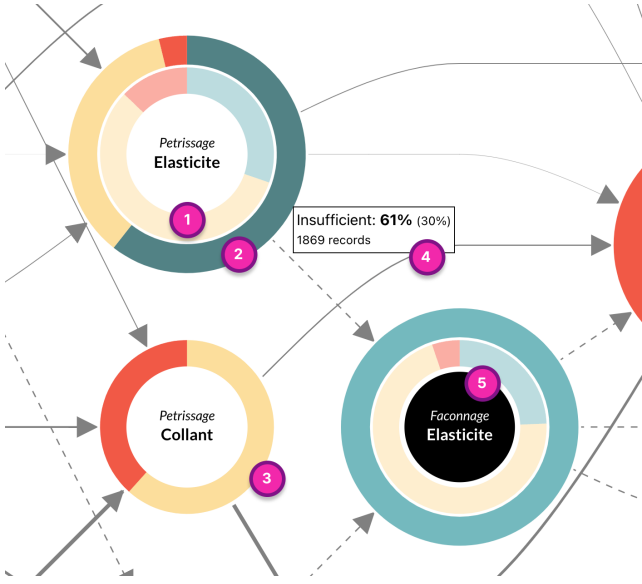
MBs can assist domain experts in narrowing their observations to only the most relevant variables in the bread-making process.

5.3 Evidence Propagation

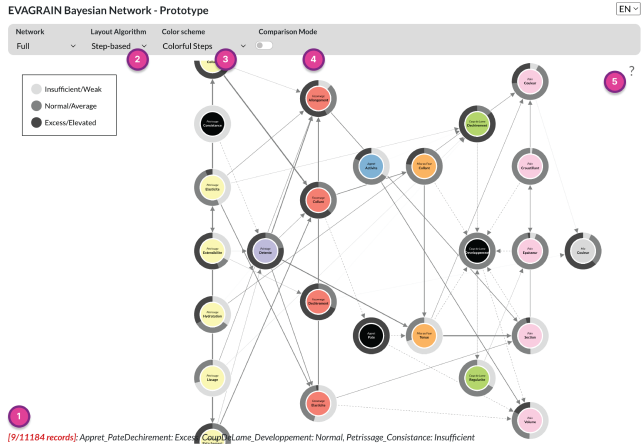
The primary interaction of the network view is evidence propagation. As suggested by the brainstorming session and related work [15, 26], we decided to omit CPTs in favor of evidence propagation (Q1). CPTs are excessively complex for experts, whereas evidence propagation allows an expert to interactively verify the relationships they expect to see in the model through indirect usage of the CPT probabilities. To propagate evidence, the user clicks on an arc in a probability ring corresponding to the desired node (e.g., “PainSection”) and sets a target evidence value (e.g. “Excess”). After selecting the desired evidence, Inference is performed and all nodes update their conditional probabilities via animation. Animation draws attention to nodes that change the most because of this propagation, due to the extra movement of their arcs (Q2). The center of the clicked node turns black to signify its ‘evidence’ status (Figure 2a.5), and the tooltip for each probability arc updates to display both the propagated and marginal probability. For example, if the probability for Pain Section - Excess jumped from 40% to 60%, the tooltip will read “Excess: 60% (40%)” (Figure 2a.4). Multiple nodes can have evidence propagated at a time. Additional evidence is added by clicking on another probability arc. A list of current evidence is displayed at the bottom of the screen.

Comparison Mode: Users may be interested in further examining a node’s probabilities before and after evidence propagation. We implemented Comparison Mode, which simultaneously shows the conditional and marginal probabilities for nodes (Figure 2a). Evidence propagation is still triggered by clicking on the target probability arc. However, rather than just animating probability rings updating, rings will simultaneously expand in diameter, revealing a ‘ghost’ copy of the original marginal probabilities. The result is a node surrounded by two rings: an inner ring of marginal probabilities (Figure 2a.1), and an outer ring of conditional probabilities for the current evidence set (Figure 2a.2). The inner ring is partially transparent to emphasize that it is secondary and only for comparison. Hovering over an outer ring arc shows the same tooltips as when comparison mode is disabled; hovering over an inner arc only shows the marginal probability. This layout is similar to that proposed in past work [7] to allow comparison between probabilities in the inner and outer ring. Nodes with two probability rings for comparisons are hereafter referred to as ‘expanded’.

Expansion Algorithms: While comparison mode is useful, some nodes may have similar probabilities before and after evidence propagation. In order to hide noise and highlight the nodes most affected by an evidence set (Q2), we implement an ‘expansion algorithm’. The expansion algorithm returns a true/false value for whether a node’s degree of change after evidence propagation is ‘significant’ enough to warrant expansion. Nodes which are not significant will still update their probabilities but remain unexpanded (Figure 2a.3). We support two expansion algorithms: (1) *Chi-Squared*: Whether the projected number of observations between the current and marginal probabilities are different according to the Chi-Squared statistical test ($p < .05$). Observations are calculated as probability * size of dataset ($n = 10,898$); and (2)



(a) Evidence Comparison in Comparison Mode. Nodes display an inner ring of marginal probabilities (1) and an outer ring of conditional probabilities (2). Nodes with insufficient change are not expanded (3). Hovering over a node displays a tooltip with current & prior probability (4). Evidence nodes have a black center (5).



(b) Prototype EVAGRAIN Visualization showing default network version with alternative settings. Unlikely evidence is propagated, triggering a warning in the evidence list (1). The network is presented using the Group Layout (2) and colorful steps color scheme (3). Comparison mode is disabled (4). The question mark (top right corner) opens the Help Menu (5).

Figure 2: Evidence Comparison in Comparison Mode (a). Prototype with alternative settings (b).

Kullback-Leibler: Calculates the relative entropy between two probability distributions to estimate the ‘distance’ between them. Its value is compared to a user-defined threshold to determine significance. The threshold is editable via a slider in the Options Bar. Actual numerical values are removed for simplicity (Q1).

5.4 Observation Count Data

Experts are interested in the amount of data informing a relation or evidence as a proxy for reliability. To support this, we include dynamic observation count data in two locations: the evidence list at the bottom left corner of the screen and the probability arc tooltips (Q4). The dataset used for observation count consists of 11,184 records. We preface the evidence list with a count of total observations in the dataset (“[11,184 records]”) (Figure 1.A6). When evidence is propagated to the network, we calculate the number of observations in the dataset matching the evidence set and present it as a fraction of total observations (e.g. “[2,234/11,184 records]”) (Figure 2b.1). If the number of observations is less than 1% of all records, we highlight the count red to indicate that inferred relationships in the BN are not reliably informed by data (Q4) (Figure 2b.1). Tooltips also show the number of observations in the dataset with the node state (e.g. “High: 33%, 3,728 records”) (Figure 1.A4). With no evidence propagated, the observation count and marginal probability are proportional since marginal probabilities reflect the underlying data. When evidence is propagated, the tooltip displays the observation count for that node state within observations matching the evidence set (Figure 2a.4). This information is useful to experts since it can highlight unreliability with network predictions (Q4).

6 Evaluation with Experts

To evaluate our platform, we conducted an informal qualitative user study with six participants. Below we describe our participants, the study protocol and our findings. Translated participant quotes are denoted by †.

Participants: We evaluated the platform in three sessions: S1 with 4 domain experts (P1-4), S2 with 1 domain expert (P5), and S3 with an external BN modeling expert (P6). All participants had over 20 years of experience. We also invited a BN expert from the original EVAGRAIN project that attended all sessions to address questions on the data and model (7 years experience). Domain experts had limited BN knowledge, mainly from previous EVAGRAIN meetings.

Protocol: Sessions S1 & S2 began with a brief explanation of BNs, followed by a demonstration of the platform and its key features. Session S3 started directly with the demonstration. Participants were then given a link to interact with the platform. They were encouraged to think aloud and share their screen to show their actions. We asked for feedback on platform features, allowing group discussion and questions. Mutual information and observation count features were not implemented for these sessions. Participants were invited to complete an optional online questionnaire to provide background information and share feedback on specific aspects of the platform. All sessions were held on Zoom, recorded on video, and each lasted around 1 hour.

Findings: Participants agreed that the platform presented a clean and visually appealing summary of the EVAGRAIN database and relationships. P3 noted: “Here, we have 10,000 rows with 34 criteria [database], ..., and we manage to summarize them in this diagram—it’s absolutely brilliant”†. Participant P5 then added: “Indeed, we had already seen what the Bayesian networks do, but when you start looking at probability tables, it’s clear that your brain quickly gets tangled up... So, it’s truly an amazing job to be able

to visualize this data while preserving the information"†; And P1: "very impressive". The strongest positive feedback was that evidence propagation was easy and visually distinctive, enabling easy probability comparisons. P5 explained how evidence propagation can be useful: *"To see the connections between the breadmaking criteria, determine which criteria to act on to correct a flaw, or preview the probability of outcomes by selecting (evidence) a few parameters"*‡. While the sessions were not intended to critique the EVAGRAIN model, participants often noted unlikely relationships and probability distributions, showing that the platform's effectiveness in BN validation. Based on participants' questionnaire responses, pie-chart probability and evidence propagation were seen the most useful and understandable features.

Domain experts preferred the group-based node layout, while the BN specialist P6 favored the link-based layout. Evidence propagation was frequently used, with all participants leaving comparison mode on. Common tasks included evidence propagation for unlikely evidence sets and setting all values of a bread-making step. Markov boundaries were rarely explored but used to identify node neighbors. The color scheme and expansion threshold settings were seldom adjusted. Experts occasionally struggled with color meanings but found the legend and tooltips helpful. Causal vs. descriptive relationships were explained verbally, and some experts seemed to understand afterward: *"But that's an observation,..., not a correlation [meaning causation]. It is only observation. In your database, when Section is in excess or in insufficiency, in that situation you observe that color or dough color or extensibility is in excess or in insufficiency, but it is not a relationship"* (P1).

Initially, experts believed the evidence propagation results depended on the order of selection (e.g., Evidence 1 then 2 vs. Evidence 2 then 1) and requested a way to select multiple nodes without updating probabilities in between. This sparked a discussion about probability calculations. Domain experts were also intrigued by improbable evidence sets and questioned how the model could make predictions with insufficient evidence. They wanted to see the underlying data in such cases. P1 noted: *"you click on Lissage, you look at insufficiency, how much observations are in this situation?"*. This led to an in-depth discussion revealing potential issues in computing some edge cases.

The BN specialist critiqued the inclusion of Markov boundaries and evidence propagation in the same platform, arguing that Markov boundaries, implemented to address Q2 (section 4), are not informative during evidence propagation and thus less useful for domain experts at this stage. However, he appreciated the evidence propagation and dynamic expansion algorithm ('significance algorithm') for examining network changes. In contrast, domain experts were confused by the expansion algorithm and used it less, but were interested in Markov boundaries as zones of influence.

7 Lessons Learned and Future Work

The visualization platform addresses most questions posed in the Design Workshops (Q1,2,4). We do not explicitly address the most contributing variable to a node (Q3), but during the validation sessions, experts used repeated evidence propagation to infer those relationships. We conveyed causality (Q5) verbally to experts.

We discuss next lessons learned from our experience:

1. Interactive Visualization Design fosters Dialogue Across Domains: During the design and evaluation of the platform, domain experts were able to reason about the model and critique it, fulfilling our initial goal. Additionally, the sessions facilitated discussions among different experts, helping them align on a shared vision of the EVAGRAIN model [4]. For example, the BN specialist identified issues with combining multiple ways to interact with BNs (e.g., evidence propagation and Markov Blankets), modeling experts found model issues by observing how domain experts interacted with and interpreted the designs, and visualization experts observed the platform in use and experts reasoning using the BN model, identifying its strengths and weaknesses to guide the next design iteration. The visualization and the design process helped the different experts better understand the implications and assumptions behind each other's work and to reach common ground [9, 10].

2. Graphical ML Models are Not Necessarily Explainable:

One of the benefits of Bayesian Networks is that they can be represented graphically, and in our case were fairly small and easy to render visually. Nevertheless, our work highlights that a graphical representation of a model does not equate an interpretable ML model. In our case, even though we created simple representations to indicate probabilities (pie-charts), our domain experts required access to raw data observations to get a better grounding of frequency and conditional probabilities. Furthermore, we had to be very careful in stressing to domain experts that directed network links represented association and not causality. It is difficult to objectively assess whether the domain experts assumed causal reasoning during investigations, although it is likely because of human tendencies towards causal thinking [28]. This question may be partially addressed by the decision to include observation count data, which emphasizes the associative nature of relations.

Future work could explore alternative visual designs beyond the node-link representation that more effectively convey association rather than causality, a comparison with other BN visualization approaches, and how to best evaluate such work in the long-run. Likely we need a mix of methodologies: using structured evaluation tasks to assess specific aspects of model understandability and validation capabilities, as done in other studies on Bayesian networks used by domain experts [12, 18, 19]; and open-ended explorations to understand the utility and insights experts gain from the interactive visualization [24]. Our ultimate goal is to allow experts to not only analyze but also correct the model. This will make evaluation even more challenging, as our work will resemble interactive visual machine learning systems, i.e., systems that learn and collaborate with users to achieve tasks mediated by interactive visual interfaces. In such systems, deciding on what evaluation metrics are important is a challenge [3]. Model explainability [29] of course is one (including conveying associations rather than causality between variables), but other factors may need to be considered, such as user trust and confidence in the model. It also remains an open question to determine if the quality of analysis and decision making does actually improve through the use of our platform, and to compare its performance to existing BN visualization tools. Finally, methods to support collaboration would be a critical step. While no expert-targeted collaborative BN platforms exist, our work shows

that validating a Bayesian Network is inherently a collaboration between model developers and domain experts.

Acknowledgments

This research was supported by DATAIA (ANR-17-CONV-0003). The authors thank the ANR EVAGRAIN project (ANR-20-CE21-0008) partners for their valuable contribution to this work, and to Pierre-Henri Wuillemin from Sorbonne University for his assistance with pyAgrum and his insightful comments and feedback.

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009